

RUNQIU BAO (潤秋 包)

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EDUCATION

Bachelor of Engineering

Sep. 2013 to Jun. 2018

Tongji University, Shanghai, P.R.China

Automotive Engineering (Electronics track), School of Automotive Studies

GPA: 4.53/5.00 Rank: 37/196(18.9%)

Exchange Program

Oct. 2017 to Apr. 2018

Technical University of Munich, Munich, Germany

Robotics, Department of Informatics

Master of Engineering

Sep. 2018 to Sep. 2020

University of Tokyo, Tokyo, Japan

Precision Engineering, School of Engineering

Research theme: Global Pose Estimation of Construction Vehicle in GNSS-denied Dynamic Environments Using Stereo Visual SLAM

PUBLICATIONS & PATENTS

Journal Publications

[1] **Runqiu Bao**, Ren Komatsu, Renato Miyagusuku, Masaki Chino, Atsushi Yamashita and Hajime Asama: "Stereo Camera Visual SLAM with Hierarchical Masking and Motion-state Classification at Outdoor Construction Sites Containing Large Dynamic Objects," Advanced Robotics, 2021. (peer-reviewed)

[\[link\]](#)

International Conferences & Symposiums

[1] **Runqiu Bao**, Ren Komatsu, Renato Miyagusuku, Masaki Chino, Atsushi Yamashita and Hajime Asama: "Cost-effective and Robust Visual Based Localization with Consumer-level Cameras at Construction Sites," Proceedings of the 2019 IEEE Global Conference on Consumer Electronics (GCCE 2019), pp. 983–985, 2019. (peer-reviewed)

[\[link\]](#)

[2] Runqiu Bao, "Feature-based Stereo Camera Ego-motion Tracking and its application at Construction Sites", the 4th Tsinghua University-The University of Tokyo Joint Multidisciplinary Symposium, 2019.

Domestic Conferences

[1] 千野 雅紀, **Runqiu Bao**, 小松 廉, Renato Miyagusuku, 山下 淳, 浅間 一: "土工現場における画像を用いた位置情報取得システムの適用可能性," 土木学会令和2年度全国大会第75回年次学術講演会講演概要集, VI-1112, pp. 1-2, September 2020.

INTERNSHIP

DJI's RoboMaster National Robot Competition in China

Dec. 2014 to Aug. 2015

Tongji II team, Tongji University

- Designing and fabricating a teleoperation robotic vehicle from scratch
- Chassis and chassis controller design for the robot vehicle

HESAI tech. Intern

Jul. 2018 to Aug. 2018

Software group, [HESAI tech.](#)

- 3D environment mapping with Lidar and IMU mounted on a robot vehicle

Image Lab Intern

2021

HUAWEI Tokyo Research Institute

- Video Deblur and Frame Interpolation with Event Camera (by Deep Learning)

Jan. 2020 to Oct.

JOB

Mujin Inc.

Now

Robotics & Computer Vision Engineer, Department of Development

- 3D object detector development: Geometry-based, contour-based, stereo camera learning-based detector algorithm design and product development.
- Detector software architecture: multi-task executor development.
- Metal piece picking robotic cell development and deployment.

Oct. 2021 until

HONORS

Awards:

Title	Organization	Year
The Second Class Prize of Mathematical Competition in Modeling (Group Award)	Tongji University	Spring, 2015
The National Second Class Prize of RoboMaster Robot Competition in China (Group Award)	DJI Innovation Company Chinese youth league	Summer, 2015
Outstanding Graduate of Tongji University	Tongji University	June, 2018
Outstanding Master's Thesis Award	Department of Precision Engineering, University of Tokyo	September, 2020

SKILLS

Languages: Chinese ~ Native, English ~ TOEFL 100, Japanese ~ JLPT N1

Programming Languages: C/C++, Python, Matlab, etc.

Familiar Algorithms and Technologies Used in Research:

- Deep learning with Pytorch
- Multi-view geometry & visual SLAM.
- Arm robot motion planning and control.